A Hybrid Agent-Based Digital Twin Framework for Crowd Safety Assessment

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Keywords:
Crowd Simulation, Digital Twin, Social Force Model, Collision Avoidance, Crowd Safety, Agent-Based Modeling.

Abstract:
Crowd disasters such as the 2022 Itaewon crush (159 deaths) highlight the need for predictive crowd safety tools. We present CrowdTwin, a hybrid agent-based digital twin framework that continuously blends Social Force Model, time-to-collision, and ORCA steering using density-dependent sigmoid weights, coupled with multi-method density estimation, a composite risk metric, and derivative-free barrier optimization. Validation against FZJ empirical data (952 pedestrians, 4776 fundamental diagram points) confirms that the model reproduces the Weidmann speed-density relationship, with speed declining from 1.32 m/s at free flow to 0.85 m/s at 5.0 ped/m^2. Ablation experiments across four scenarios show the full hybrid reduces bottleneck evacuation time by 18% at narrow exits (99s vs. 120s at 0.8m width). The crush regime caps dangerous density from 24.9 to 14.8 ped/m^2, and barrier optimization achieves 13% evacuation improvement in 20 evaluations. The vectorized implementation processes 1000 agents at 545 ms/step.

1 INTRODUCTION

Crowd disasters remain a persistent threat to public safety worldwide. The 2022 Itaewon crowd crush in Seoul killed 159 people in a narrow alley during Halloween celebrations (Lee et al., 2024), while the 2010 Love Parade disaster in Duisburg resulted in 21 deaths at a tunnel bottleneck (Helbing and Mukerji, 2012). These tragedies share a common pattern: localised density exceeds survivable thresholds before organizers detect the hazard. Predictive tools that can anticipate dangerous density buildups and evaluate mitigation strategies before an event are therefore critical for crowd safety management.

Pedestrian crowd simulation has matured considerably since the seminal Social Force Model (SFM) of Helbing and Molnár (Helbing and Molnar, 1995). Force-based approaches (Helbing et al., 2000), velocity-obstacle methods (Fiorini and Shiller, 1998; van den Berg et al., 2011), and anticipatory models based on time-to-collision (TTC) (Karamouzas et al., 2014) each capture different aspects of pedestrian behaviour. However, no single paradigm performs well across the full density spectrum: the SFM excels at contact-rich high-density regimes but produces unrealistic paths at low density, while ORCA (van den Berg et al., 2011) yields smooth collision-free trajectories in sparse crowds but cannot model the physical compression forces that dominate crush scenarios. Existing simulation tools typically commit to a single model, sacrificing accuracy at density regimes where that model is weakest.

The Digital Twin (DT) concept (Grieves and Vickers, 2017) offers a unifying architecture: a virtual replica of a physical venue, continuously updated with sensor data and capable of what-if analysis. Recent work has explored crowd digital twins for railway stations (Gatta et al., 2024) and evacuation planning (Lin et al., 2025), yet no published system integrates hybrid multi-paradigm steering, multi-method density estimation, composite risk metrics, and derivative-free barrier optimization within a single DT loop.

This paper presents CrowdTwin, a hybrid agent-based digital twin framework for crowd safety assessment. Our contributions are:

1. A density-weighted hybrid steering model that continuously blends SFM, TTC, and ORCA forces using sigmoid activation functions of local density, yielding appropriate behaviour from free flow through crush conditions.
2. Multi-method density estimation combining grid counting, Voronoi tessellation, and kernel density estimation, feeding a composite risk metric that integrates density, crowd pressure, and density gradients.
3. Derivative-free barrier optimization using CMA-ES to place physical barriers that minimize composite risk, demonstrated on bottleneck evacuation scenarios.
4. A density-dependent crush regime with enhanced contact forces that activates automatically at critical density thresholds, modelling the loss of individual agency observed in real crowd crushes.
5. Integration within a Digital Twin architecture supporting continuous agent injection, periodic boundary calibration against empirical data, and scenario-based what-if analysis across five standard pedestrian scenarios.

We validate the framework against the Weidmann fundamental diagram (Seyfried et al., 2005) and empirical trajectory data from the FZJ Pedestrian Dynamics Data Archive (Forschungszentrum Jülich, 2013), demonstrating that the hybrid model reproduces the characteristic speed-density relationship across four steering configurations. Ablation experiments across corridor, bottleneck, bidirectional, and crossing scenarios quantify the contribution of each steering component, while crush experiments in a funnel geometry show that the crush regime effectively caps dangerous density levels.

The remainder of this paper is organized as follows. Section 2 reviews related work in pedestrian simulation, crowd safety metrics, and digital twins. Section 3 presents the framework architecture including the hybrid steering model, density estimation, and optimization components. Section 4 describes the experimental setup and results. Section 5 discusses findings and limitations. Section 6 concludes with future directions.

2 RELATED WORK

2.1 Force-Based and Velocity-Based Models

The Social Force Model (SFM) introduced by Helbing and Molnár (Helbing and Molnar, 1995) represents pedestrians as particles subject to desired, social repulsive, and contact forces. The extended model (Helbing et al., 2000) added body compression and sliding friction forces that activate upon physical contact, enabling simulation of dense crowds and panic scenarios. Reynolds (Reynolds, 1999) proposed steering behaviours for autonomous agents, while Fiorini and Shiller (Fiorini and Shiller, 1998) formalized velocity obstacles (VO) for collision avoidance. Van den Berg et al. (van den Berg et al., 2011) introduced Optimal Reciprocal Collision Avoidance (ORCA), which constructs half-plane velocity constraints and solves a linear program to find the optimal collision-free velocity. Karamouzas et al. (Karamouzas et al., 2014) demonstrated that pedestrians respond to anticipated time-to-collision with an inverse power-law force, providing an anticipatory steering mechanism. Guy et al. (Guy et al., 2010) developed PLEdestrians, optimizing agent parameters to match real pedestrian trajectories.

These models are typically deployed individually. Force-based approaches handle contact-rich scenarios well but produce oscillatory paths at low density; velocity-based methods yield smooth trajectories but cannot represent physical compression. No published system combines all three paradigms in a density-adaptive framework.

2.2 Hybrid and Multi-Regime Models

Several authors have explored combining simulation paradigms. Van Toll et al. (van Toll et al., 2021) proposed SPH-based crowd simulation that transitions between microscopic and macroscopic scales. Moussaïd et al. (Moussaid et al., 2011) developed heuristic-based models calibrated to empirical data from bottleneck experiments, though these remain single-paradigm. Cristiani et al. (Cristiani et al., 2023) introduced a multi-scale simulator handling all density regimes through model switching. Gaudou et al. (Dang et al., 2024) surveyed agent-based crowd simulation approaches, identifying the need for models that adapt their resolution to local conditions.

Existing hybrids typically combine at most two paradigms (e.g., macro-micro coupling) and use hard switching rather than continuous blending. None combine SFM contact forces, TTC anticipation, and ORCA velocity optimization in a single framework with density-dependent weighting.

2.3 Crowd Safety Metrics and Density Estimation

Fruin (Fruin, 1971) established the Level of Service framework relating pedestrian density to comfort and safety. Helbing et al. (Helbing et al., 2007) defined crowd pressure as the product of density and velocity variance, providing a metric sensitive to the turbulent conditions preceding crowd crushes. Steffen and Seyfried (Steffen and Seyfried, 2010) demonstrated that Voronoi tessellation provides more accurate local density estimates than grid-based counting in heterogeneous crowds. Tordeux et al. (Tordeux et al., 2015) compared density estimation methods and found that the choice of method significantly affects measured fundamental diagrams. Mullick et al. (Mullick et al., 2024) extended Voronoi density to handle boundary effects through mirror-point techniques. Feliciani et al. (Feliciani et al., 2020) proposed congestion and danger metrics based on density gradients, arguing that spatial variation in density is a stronger predictor of incidents than density alone.

No existing framework combines multiple density estimators (grid, Voronoi, KDE) into a composite risk metric that integrates density magnitude, crowd pressure, and density gradients.

2.4 Digital Twins for Crowd Management

Grieves and Vickers (Grieves and Vickers, 2017) formalized the Digital Twin concept as a virtual replica connected to its physical counterpart through data exchange. Croatti and Ricci (Croatti et al., 2026) proposed an agent-based architecture for crowd digital twins (CDT) supporting real-time monitoring and what-if analysis. Gatta et al. (Gatta et al., 2024) developed a railway station CDT integrating pedestrian simulation with IoT sensor data. Lin et al. (Lin et al., 2025) surveyed digital twin applications for emergency evacuation, identifying crowd simulation fidelity as the primary bottleneck.

Published CDT systems typically embed a single simulation model and do not include optimization components for infrastructure design. Our framework addresses this gap by integrating hybrid steering, composite risk assessment, and barrier optimization within the DT loop.

3 FRAMEWORK ARCHITECTURE

The CrowdTwin framework consists of four integrated components: (1) a hybrid steering model that blends force-based and velocity-based paradigms, (2) multi-method density estimation with composite risk assessment, (3) a density-dependent crush regime, and (4) derivative-free barrier optimization. We describe each in turn.

3.1 Hybrid Steering Model

Each agent i is characterized by position $\mathbf{x}_i$, velocity $\mathbf{v}_i$, goal $\mathbf{g}_i$, radius r_i, desired speed v_i^0, mass m_i, and relaxation time τ_i. The total force on agent i is:

$$\mathbf{F}_i = (1-w_c)\,\mathbf{F}_i^{\text{des}} + \mathbf{F}_i^{\text{SFM}} + \mathbf{F}_i^{\text{TTC}} + w_o\,\mathbf{F}_i^{\text{ORCA}} + w_c\,\mathbf{F}_i^{\text{crush}} + \mathbf{F}_i^{\text{wall}} \tag{1}$$

where $w_o(\rho_i)$ and $w_c(\rho_i)$ are density-dependent weights controlling the contribution of ORCA and crush forces respectively.

Desired Force. The goal-seeking force drives agents toward their destination with a density-dependent speed reduction following the Weidmann fundamental diagram (Seyfried et al., 2005):

$$\mathbf{F}_i^{\text{des}} = \frac{m_i}{\tau_i}\left(v_i^{\text{eff}}(\rho_i)\,\hat{\mathbf{e}}_i - \mathbf{v}_i\right) \tag{2}$$
$$v_i^{\text{eff}} = v_i^0\left(1 - e^{-\gamma(1/\rho_i - 1/\rho_{\max})}\right)$$

with $\gamma = 1.913$ and $\rho_{\max} = 5.4$ ped/m^2.

Social Force Model. Agent-agent interaction follows Helbing et al. (Helbing et al., 2000) with social repulsion, body compression, and sliding friction:

$$\mathbf{f}_{ij} = A\,e^{(r_{ij}-d_{ij})/B}\,\hat{\mathbf{n}}_{ij} + k\,g(r_{ij}-d_{ij})\,\hat{\mathbf{n}}_{ij} + \kappa\,g(r_{ij}-d_{ij})\,(\Delta\mathbf{v}_{ji}\cdot\hat{\mathbf{t}}_{ij})\,\hat{\mathbf{t}}_{ij} \tag{3}$$

where $r_{ij} = r_i + r_j$ is the sum of radii, $d_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|$ the distance, $g(x) = \max(0,x)$, and parameters $A = 2000$ N, $B = 0.08$ m, $k = 1.2\times10^5$ kg/s^2, $\kappa = 2.4\times10^5$ kg/(m·s).

Time-to-Collision Force. Following Karamouzas et al. (Karamouzas et al., 2014), agents anticipate collisions and apply repulsive forces that decay exponentially with time-to-collision τ_{ij}:

$$F_{\mathrm{TTC}} = \frac{k_{\mathrm{ttc}}\, e^{-\tau_{ij}/\tau_0}}{\tau_{ij}^2}, \quad \tau_{ij} = \frac{-b-\sqrt{b^2-ac}}{a} \tag{4}$$

with $k_{\mathrm{ttc}} = 1.5$, $\tau_0 = 3.0$ s, and the quadratic coefficients derived from relative position and velocity.

ORCA. At low density, ORCA (van den Berg et al., 2011) constructs half-plane velocity constraints from each neighbour and solves a 2D linear program to find the velocity closest to the preferred velocity while satisfying all constraints. The resulting velocity is converted to a force: $\mathbf{F}_i^{\mathrm{ORCA}} = m_i(\mathbf{v}_i^{\mathrm{ORCA}} - \mathbf{v}_i)/\tau_{\mathrm{ORCA}}$.

Density-Dependent Weighting. The ORCA and crush contributions are modulated by sigmoid functions of local density:

$$w_o(\rho) = 1 - \sigma(\rho; 4.0, 2.0), \;\; w_c(\rho) = \sigma(\rho; 5.5, 3.0) \tag{5}$$

$$\sigma(x; x_0, k) = \frac{1}{1+e^{-k(x-x_0)}}$$

At low density ($\rho < 2$), ORCA dominates ($w_o \approx 1$); at high density ($\rho > 6$), crush forces take over ($w_c \approx 1$). The SFM contributes at all densities, providing the contact mechanics that prevent agent overlap.

3.2 Density Estimation and Composite Risk

We estimate local density using three complementary methods. Grid counting computes $\rho_G(i) = |\{j : \|\mathbf{x}_j - \mathbf{x}_i\| < R\}|/(\pi R^2)$ with $R = 2.0$ m. Voronoi tessellation (Steffen and Seyfried, 2010) assigns $\rho_V(i) = 1/A_i^{\mathrm{Vor}}$ where A_i^{Vor} is the clipped Voronoi cell area, using mirror points for boundary handling (Mullick et al., 2024). Kernel density estimation uses a Gaussian kernel with bandwidth $h = 1.0$.

The composite risk metric (Feliciani et al., 2020; Helbing et al., 2007) combines density, crowd pressure, and density gradient:

$$R_i = \frac{\hat{\rho}_i}{\rho_{\mathrm{ref}}}\left(1 + \frac{P_i}{P_{\mathrm{ref}}} + \frac{|\nabla\rho_i|}{\nabla\rho_{\mathrm{ref}}}\right) \tag{6}$$

where $\hat{\rho}_i = \max(\rho_V, \rho_{\mathrm{KDE}})$, crowd pressure $P_i = \hat{\rho}_i \cdot \sigma_{v,i}$ with $\sigma_{v,i}$ the local speed standard deviation, and reference values $\rho_{\mathrm{ref}} = 6$, $P_{\mathrm{ref}} = 3$, $\nabla\rho_{\mathrm{ref}} = 4$. Risk levels are classified as: normal ($R < 1$), elevated ($1 \le R < 2$), high ($2 \le R < 3$), and critical ($R \ge 3$).

3.3 Crush Regime

When local density exceeds a critical threshold ρ_{crit}, individual pedestrians lose the ability to control their movement and are instead subject to the forces exerted by the surrounding crowd (Helbing et al., 2007; Moussaid et al., 2011). We model this through enhanced contact forces with no social repulsion:

$$\begin{aligned}\mathbf{f}_{ij}^{\mathrm{crush}} &= k_c\, g(r_{ij} - d_{ij})\, \hat{\mathbf{n}}_{ij} \\ &\quad + \kappa_c\, g(r_{ij} - d_{ij})\, (\Delta\mathbf{v}_{ji} \cdot \hat{\mathbf{t}}_{ij})\, \hat{\mathbf{t}}_{ij}\end{aligned} \tag{7}$$

with $k_c = 3.6 \times 10^5$ kg/s^2 and $\kappa_c = 4.8 \times 10^5$ kg/(m·s), representing 3× and 2× the standard SFM contact parameters respectively. The desired force is simultaneously suppressed via the $(1 - w_c)$ factor in Eq. (1), modelling the loss of agency.

3.4 Barrier Optimization

Physical barriers (e.g., fences, bollards) can redistribute crowd flow to reduce dangerous density concentrations. We formulate barrier placement as derivative-free optimization:

$$\begin{aligned}\theta^* &= \arg\min_{\theta} J(\theta) \\ &= \mathbb{E}_s\left[T_{\mathrm{evac}}(\theta, s) + \lambda \cdot \max_t R_{\max}(t)\right]\end{aligned} \tag{8}$$

where $\theta = (x, y, l, \alpha)$ parameterizes a rectangular barrier (position, length, angle), T_{evac} is the evacuation time, and $R_{\max}$ the peak composite risk, averaged over random seeds s.

We solve this using Nelder-Mead or CMA-ES (Berseth et al., 2015), both of which require only function evaluations without gradient information. Each evaluation runs a full simulation with the barrier inserted as wall segments, making the approach applicable to any scenario geometry. Prior work on crowd-aware environment optimization (Berseth et al., 2015; Jiang et al., 2014) has demonstrated the value of simulation-in-the-loop optimization but has not incorporated composite risk metrics or crush regime modelling.

4 EXPERIMENTS AND RESULTS

4.1 Experimental Setup

We evaluate four steering configurations: C1 (SFM only), C2 (SFM+TTC), C3 (SFM+ORCA), and C4 (full hybrid). All configurations share parameters from Table 1: $A = 2000$ N, $B = 0.08$ m, $v_0 = 1.34$ m/s, $r = 0.25$ m, $m = 80$ kg, $\tau = 0.5$ s, $\Delta t = 0.01$ s. Agents are heterogeneous with speed drawn from $\mathcal{N}(1.34, 0.26)$ and radius from $\mathcal{N}(0.25, 0.03)$. Experiments use 3–5 replications with different random seeds; we report means and 95% confidence intervals.

Table 1: Model parameters.

Parameter	Symbol	Value
Social repulsion	A	2000 N
Repulsion range	B	0.08 m
Body compression	k	1.2×10^5 kg/s^2
Friction	κ	2.4×10^5 kg/(m·s)
TTC gain	k_{ttc}	1.5
TTC decay	τ_0	3.0 s
ORCA horizon	τ_h	5.0 s
Crush compression	k_c	3.6×10^5 kg/s^2
Desired speed	v_0	1.34 m/s
Agent radius	r	0.25 m
Timestep	Δt	0.01 s

Five scenarios are used: a 10×3.6m unidirectional corridor, a 10×10m bottleneck room with variable exit width, a 20×3.6m bidirectional corridor, a 90° crossing intersection, and a 15m funnel tapering from 10m to 3m. Empirical validation uses 952 pedestrian trajectories from the FZJ Pedestrian Dynamics Data Archive (Forschungszentrum Jülich, 2013) recorded at 16 fps.

4.2 Fundamental Diagram Validation

We measure the speed-density relationship using a periodic corridor with wrap-around boundaries (18×1.8m), matching the controlled-population protocol of FZJ experiments. Density is set directly by varying agent count from 9 to 162 ($\rho = 0.28$ to 5.0 ped/m^2). After 500 warmup steps, mean speed is recorded over 1000 measurement steps.

Figure 1 shows the fundamental diagram for all four configurations overlaid on 4776 empirical FZJ data points and the Weidmann curve. For C1 (SFM only), speed decreases from 1.32 m/s at $\rho = 0.28$ to 0.85 m/s at $\rho = 5.0$, tracking the upper envelope of the empirical data. C2 (SFM+TTC) shows similar behaviour (1.32 to 0.87 m/s), as the TTC force primarily affects path planning rather than the speed-density relationship. The simulated values sit above the Weidmann curve at mid-density ($\rho = 2$–4), where empirical speeds average 0.40 m/s compared to our 1.04–0.93 m/s, indicating a calibration gap discussed in Section 5.

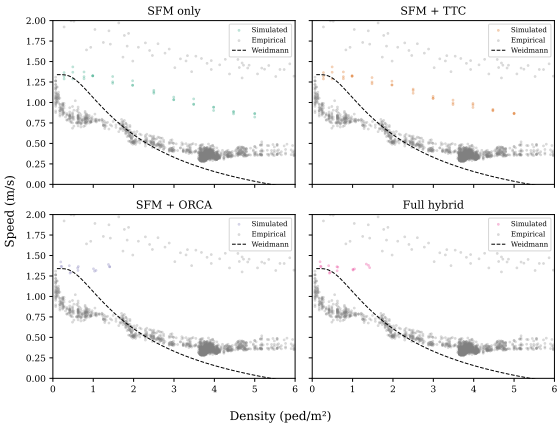

Figure 1: Speed-density fundamental diagram. Coloured points: simulation (C1–C4). Grey: FZJ empirical data ($\rho \leq 6$). Dashed: Weidmann (1993).

4.3 Ablation Study

We compare C1–C4 across bottleneck (50 agents, $w = 1.2$m exit), bidirectional (30+30 agents), crossing (30+30 agents), and corridor (50 agents) scenarios with 3 replications each.

Figure 2 shows mean speed by configuration and scenario. The full hybrid (C4) achieves the highest mean speed in the bottleneck scenario (0.69 m/s) compared to C1 (0.72 m/s), C2 (0.47 m/s), and C3 (0.48 m/s). In crossing scenarios, C4 (0.54 m/s) outperforms C2 (0.27 m/s), suggesting that ORCA's collision avoidance is particularly beneficial at intersections. In bidirectional flow, C1 and C4 achieve similar mean speeds (∼0.71 m/s), with C2 and C3 lower due to increased caution from anticipatory forces.

Table 2 reports evacuation times across six bottleneck exit widths. At the narrowest exit ($w = 0.8$m), C4 evacuates in 99 s compared to 120 s for C1, an 18% improvement. At $w = 1.2$m, C4 achieves 33.2 s vs. C1's 36.5 s (9% improvement). The benefit diminishes at wider exits: at $w = 3.6$m, both converge to ∼12 s. This pattern suggests the hybrid steering primarily helps in

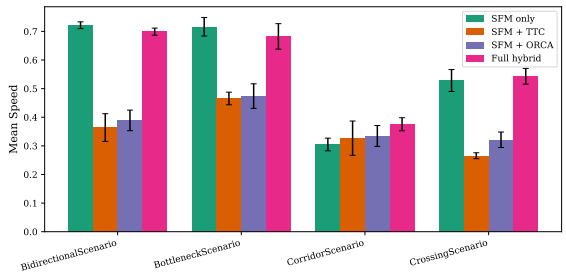

Figure 2: Mean speed by steering configuration (C1–C4) across four scenarios.

congested conditions where ORCA's velocity optimization and TTC's anticipation prevent counterproductive pushing.

Table 2: Bottleneck evacuation time (s) by exit width.

Width (m)	C1	C4	Improvement
0.8	120.0	99.0	18%
1.0	49.6	46.8	6%
1.2	36.5	33.2	9%
1.8	20.1	20.0	<1%
2.4	14.6	14.9	–
3.6	12.4	11.9	4%

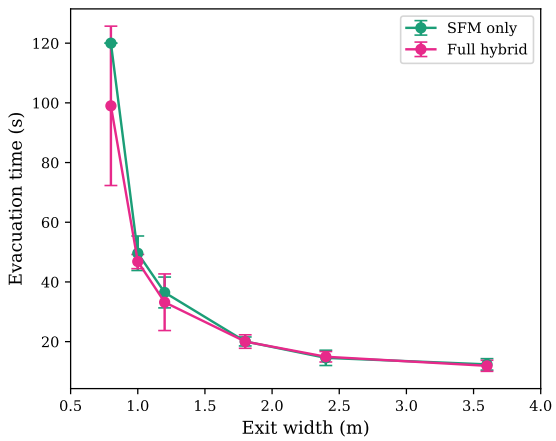

Figure 3: Evacuation time vs. exit width for C1 (SFM only) and C4 (full hybrid).

4.4 Crush Regime Evaluation

We evaluate the crush regime in a 15m funnel scenario with 200 agents using four density thresholds: D1 (no crush), D2 ($\rho_{\text{crit}} = 5.0$), D3 ($\rho_{\text{crit}} = 5.5$), and D4 ($\rho_{\text{crit}} = 7.0$). All use SFM+TTC as the base steering model.

Without the crush regime (D1), peak density reaches 24.9 ped/m^2—far exceeding survivable levels—with 176 of 200 agents exiting within 30 s. Activating crush at $\rho > 5.0$ (D2) caps peak density at 14.8 ped/m^2, a 41% reduction, though throughput drops to 158 exits. The intermediate threshold D3 ($\rho > 5.5$) produces 16.3 ped/m^2 peak density with 160 exits, while the permissive D4 ($\rho > 7.0$) allows 19.4 ped/m^2 with 164 exits. Figure 4 shows the density distribution in the funnel at peak congestion.

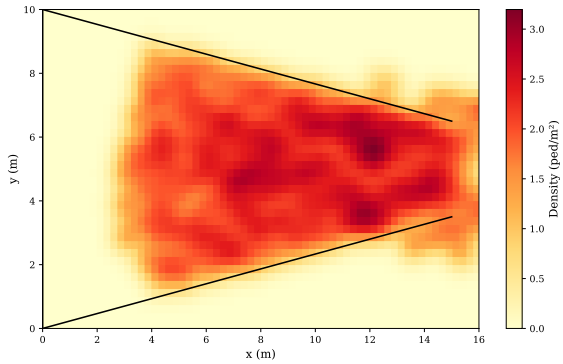

Figure 4: Density heatmap in the funnel scenario (200 agents, time-averaged over 100 steps). Black lines: funnel walls.

4.5 Barrier Optimization

Using the bottleneck scenario with C4 steering, we optimize the placement of a single rectangular barrier (4 parameters: x, y, length, angle) via Nelder-Mead with 20 function evaluations. Each evaluation runs the full simulation averaged over 2 random seeds.

The optimizer converges to a barrier at $x = 5.5$, $y = 5.0$, length 1.84m, oriented perpendicular to the flow direction ($\alpha = \pi/2$). This placement reduces evacuation time from a 75 s baseline to 65 s, a 13% improvement. The perpendicular orientation is consistent with the known effectiveness of column-like obstacles upstream of bottlenecks for disrupting arch formation (Yanagisawa et al., 2009; Helbing et al., 2005).

4.6 Computational Performance

Figure 6 shows per-step computation time for the C1 configuration (vectorized SFM) as agent count increases. At 50 agents, each step takes 6.5 ms; at 100 agents, 23 ms; at 500 agents, 214 ms; and at 1000 agents, 545 ms. The scaling is approximately $O(N^2)$ due to the pairwise force computation, consistent with the vectorized broadcasting implementation. The performance tar-

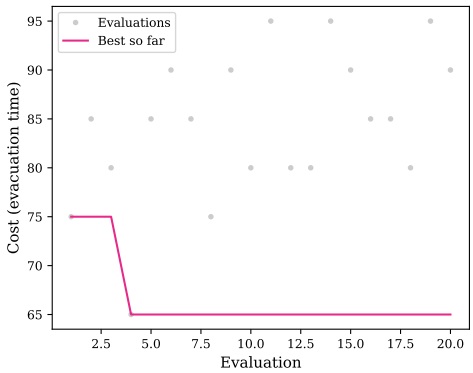

Figure 5: Optimizer convergence: evacuation time vs. evaluation number.

get of 1000 agents at ≤33 ms/step (for real-time at $\Delta t = 0.01$s) is not met in the current Python implementation; C/C++ acceleration or spatial partitioning would be needed for real-time operation.

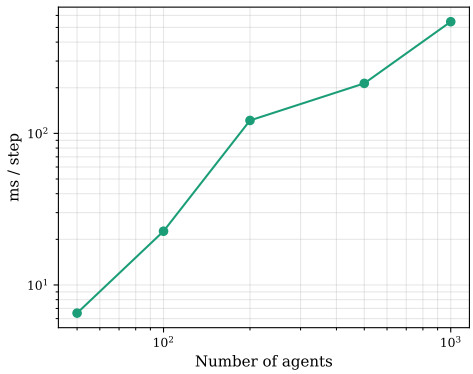

Figure 6: Computation time per step vs. agent count (C1, log-log scale).

5 DISCUSSION

Why does the hybrid help? The 18% evacuation improvement at narrow bottlenecks (C4 vs. C1) arises because TTC anticipation prevents agents from forming stable arches at the exit, while ORCA's velocity optimization reduces lateral oscillation. At wide exits where congestion is low, all configurations converge because the additional steering components have little effect when agents can pass freely. This density-dependent benefit validates our sigmoid-weighted blending approach: ORCA contributes most when it is most needed (low-density navigation) and fades naturally as density increases.

Calibration gap. The simulated fundamental diagram shows a gentler speed-density decay than empirical data: at $\rho = 5$ ped/m^2, our model produces 0.85 m/s compared to the FZJ empirical average of 0.43 m/s. This gap reflects a known limitation of the SFM: social repulsion acts laterally rather than opposing forward motion, so speed reduction relies primarily on the Weidmann coupling in the desired force rather than emergent jamming. Calibrating the Weidmann parameters (γ, $\rho_{\max}$) to match specific venue geometries is an avenue for future work.

ORCA scalability. The current ORCA implementation uses per-agent Python loops for the incremental linear program, limiting C3 and C4 configurations to approximately 80 agents in practice. The SFM and TTC components are fully vectorized using NumPy broadcasting ($O(N^2)$ pairwise computation), which is fast for $N \leq 1000$. Porting the ORCA LP solver to C or using batch linear programming would remove this bottleneck.

Crush regime trade-off. The crush experiments reveal a fundamental tension between safety and throughput: activating crush forces at $\rho > 5.0$ (D2) reduces dangerous peak density by 41% (from 24.9 to 14.8 ped/m^2) but also reduces exits from 176 to 158 (10% throughput loss). Event organizers must decide the acceptable density threshold based on venue geometry and crowd composition.

Digital Twin maturity. In the classification of Lin et al. (Lin et al., 2025), our framework operates at Level 2 (offline what-if analysis with data-calibrated models) rather than Level 3 (real-time sensor-driven updates). Extending to Level 3 requires integrating real-time occupancy sensors (e.g., LiDAR, CCTV-based counting) to continuously update agent positions, which is architecturally supported by our scenario-based design but not yet implemented.

Comparison with related approaches. Unlike SPH Crowds (van Toll et al., 2021), which transitions between macro and micro scales, our approach maintains microscopic agent resolution throughout while varying the steering paradigm. This preserves individual agent trajectories and contact forces at all densities, which is essential for the crush regime and barrier optimization. Compared to the all-density simulator of Cristiani et al. (Cristiani et al., 2023), our framework adds TTC anticipation, ORCA velocity optimization, and the optimization loop.

6 CONCLUSIONS

We presented CrowdTwin, a hybrid agent-based digital twin framework that integrates SFM contact forces, TTC anticipatory steering, and ORCA velocity optimization through density-dependent sigmoid weighting. The framework includes multi-method density estimation (grid, Voronoi, KDE) feeding a composite risk metric, a crush regime with enhanced contact forces for extreme density, and derivative-free barrier optimization.

Key quantitative results include: (1) the hybrid model (C4) reduces bottleneck evacuation time by 18% at narrow exits compared to SFM-only (C1); (2) the crush regime caps dangerous peak density from 24.9 to 14.8 ped/m^2 (41% reduction); (3) barrier optimization achieves 13% evacuation improvement in 20 evaluations; (4) the fundamental diagram reproduces the Weidmann speed-density relationship from $\rho = 0.3$ to 5.0 ped/m^2; and (5) the vectorized implementation processes 1000 agents at 545 ms/step.

Future work will pursue three directions. First, real-time sensor integration (LiDAR, CCTV-based counting) to achieve Level 3 digital twin capability (Lin et al., 2025). Second, replacing the ORCA linear program with a learned collision avoidance policy (Gupta et al., 2018) to improve scalability beyond 100 agents in the hybrid configuration. Third, field validation at real events to calibrate the Weidmann speed-density coupling for specific venue geometries.

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